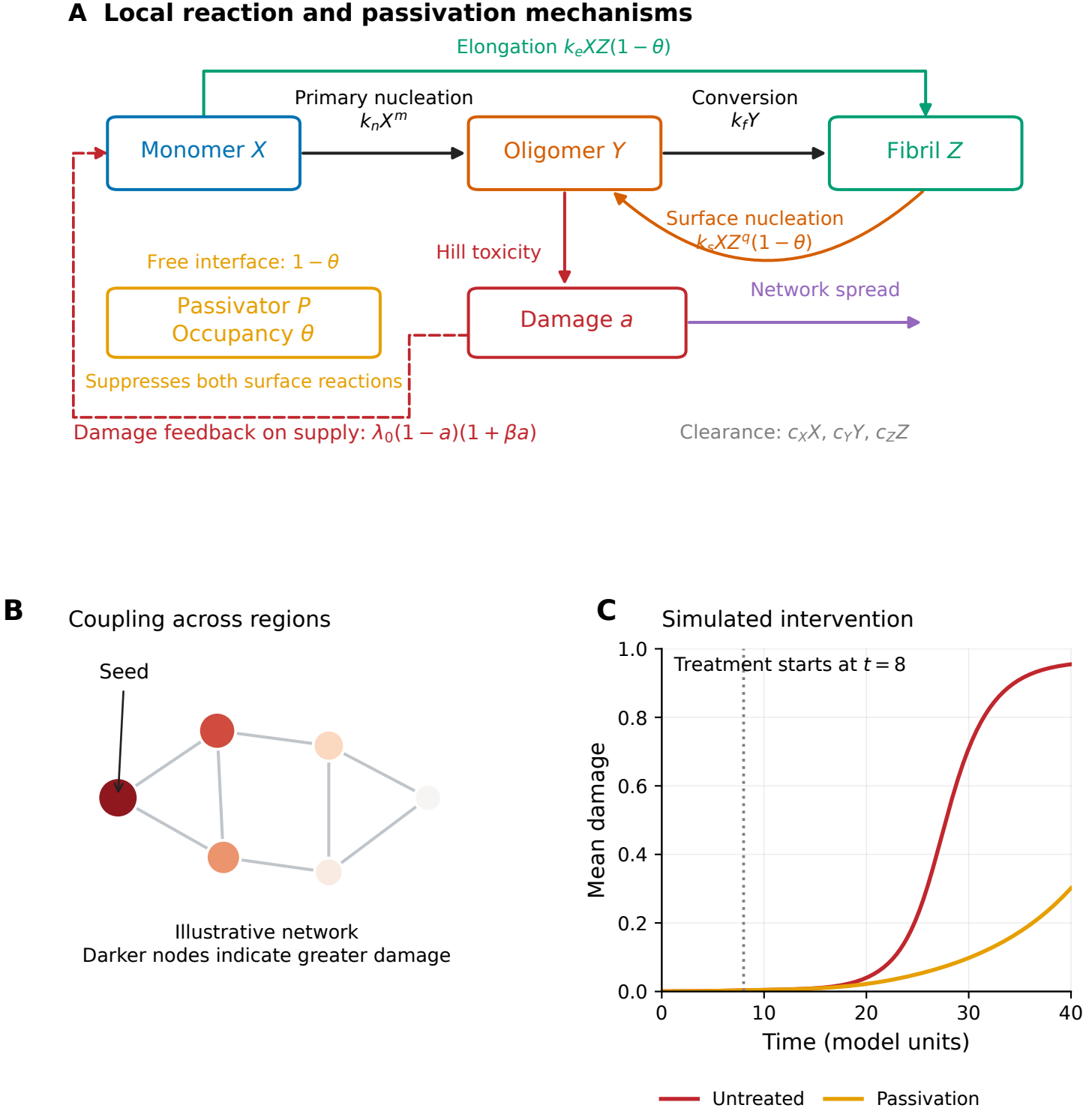
A Local reaction and passivation mechanisms
Elongation $k_e XZ(1-\theta)$
Primary nucleation $k_n X^m$
Conversion $k_f Y$
Monomer X
Oligomer Y
Fibril Z
Surface nucleation $k_s XZ^q(1-\theta)$
Hill toxicity
Free interface: $1-\theta$
Passivator P
Occupancy θ
Damage a
Network spread
Suppresses both surface reactions
Damage feedback on supply: $\lambda_0(1-a)(1+\beta a)$
Clearance: $c_X X$, $c_Y Y$, $c_Z Z$
B
Coupling across regions
Seed
Illustrative network
Darker nodes indicate greater damage
C
Simulated intervention
Treatment starts at $t = 8$
Mean damage
1.0
0.8
0.6
0.4
0.2
0.0
0
10
20
30
40
Time (model units)
Untreated
Passivation